

# Deepak Pathak

Guwahati, Assam, India

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## EDUCATION

### Tezpur University

B.Tech in Computer Science and Engineering

2023 – Present

CGPA: 7.30

### IIT Madras

BS in Data Science

2023 – Present

CGPA: 6.95

## TECHNICAL SKILLS

- **Languages:** Python, C/C++, Kotlin, Java, JavaScript, SQL
- **Web & Frameworks:** Flask, FastAPI, Spring Boot, VueJS, Jinja2, Bootstrap, SQLAlchemy
- **Tools & Databases:** SQLite, Redis, Celery, Pandas, NumPy, LangChain
- **Core CS & Networking:** TCP/UDP Socket Programming, POSIX Threads, Data Structures, Algorithms, DBMS

## EXPERIENCE

### DSeT Consulting

Jan 2026 – Present

*Software Engineering Intern*

- Developed an automated billing system using Python to streamline invoicing and payment follow-ups.
- Implemented voice call reminders and automated email notifications utilizing Twilio APIs.
- Built auto-invoice update features providing real-time billing tracking.
- Integrated Azure AI services to facilitate intelligent communication workflows.

## PROJECTS

### Placement Portal Application (PPA) | *Flask, VueJS, Redis, Celery, SQLite*

- Engineered a role-based recruitment web application with distinct access controls for Admins, Companies, and Students using JWT/Flask security.
- Implemented background batch jobs utilizing Redis and Celery to automate daily application reminders via Google Chat webhooks and generate monthly HTML activity reports.
- Optimized API response times by engineering a caching system and developed a unified user model with programmatically generated databases.

### Concurrent Network Architecture | *C, TCP/UDP Sockets, POSIX Threads*

- Developed concurrent server applications utilizing POSIX threads and TCP sockets to manage multiple client connections and ensure thread-safe operations.
- Built a real-time group chatbox with automated, timestamped message logging, alongside a multi-client inventory system for atomic stock tracking.
- Implemented a UDP-based scientific calculator, establishing reliable client-server data transfer and utilizing Wireshark for continuous packet loss detection.

### Hospital Management System | *Python, Flask, SQLAlchemy, CSS*

- Developed a full-stack application featuring multi-role authentication for Admins, Doctors, and Patients.
- Designed robust REST APIs, normalized database schemas, and secure session management.
- Built a responsive frontend utilizing Jinja2 and CSS to ensure efficient and secure data handling.

## ACHIEVEMENTS & INTERESTS

- **Achievements:** Winner, Interhostel Cricket Tournament (2025); Winner, Interhostel Kabaddi Tournament (2024).
- **Interests:** AI-powered application development, 3D printing & product design, team sports, problem-solving.